Parallel Computing

CHS 3254 & IT 3243 / Practical

Mr. I. Hariharan
Lecturer (Probationary)
Department of Physical Science,
Faculty of Applied Science,
University of Vavuniya.

Course Outline and ILO

CSH 3254

- Utilize modern parallel computing languages to implement given parallel algorithms.
- Parallel implementation of the above concepts using Openmp/MPI/CUDA.

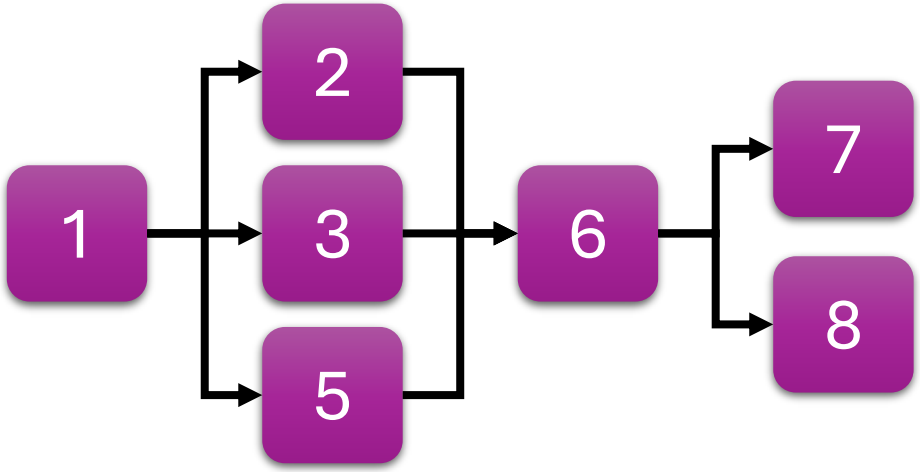
IT 3243

- Build parallel programming using message passing interface.
- Implementation of parallelized algorithms using message passing interface technique.

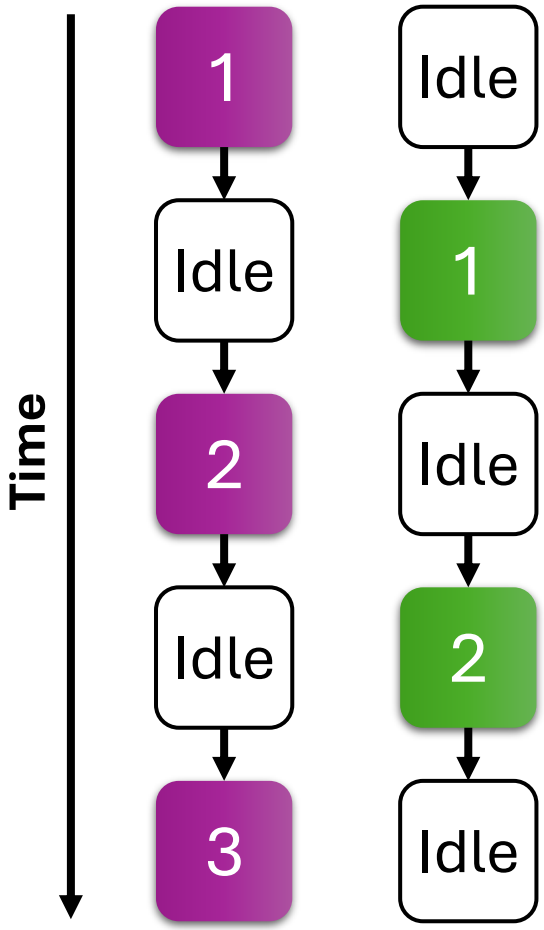
Serial VS Parallel

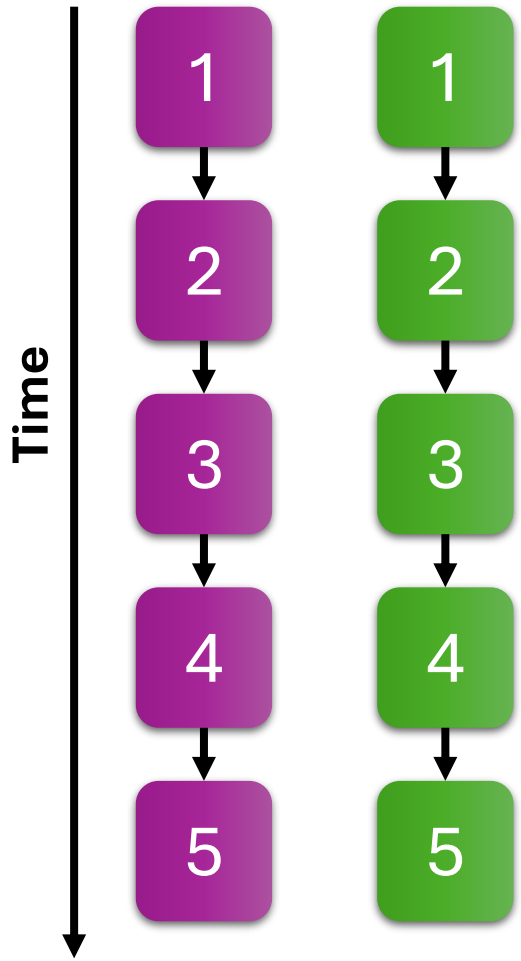
Serial processing is a method of executing tasks **one after another in a sequential order**, where each operation must be completed before the next one begins.

Parallel processing is a computational approach where **multiple tasks or instructions are executed simultaneously** by dividing the workload across multiple processors, CPU cores, or machines.



Multitasking VS Parallelism





Data Parallelism & Task Parallelism

The same task/operation is performed on different chunks of data at the same time.

Different tasks (functions) run at the same time, possibly on the same or different data.

1000 papers

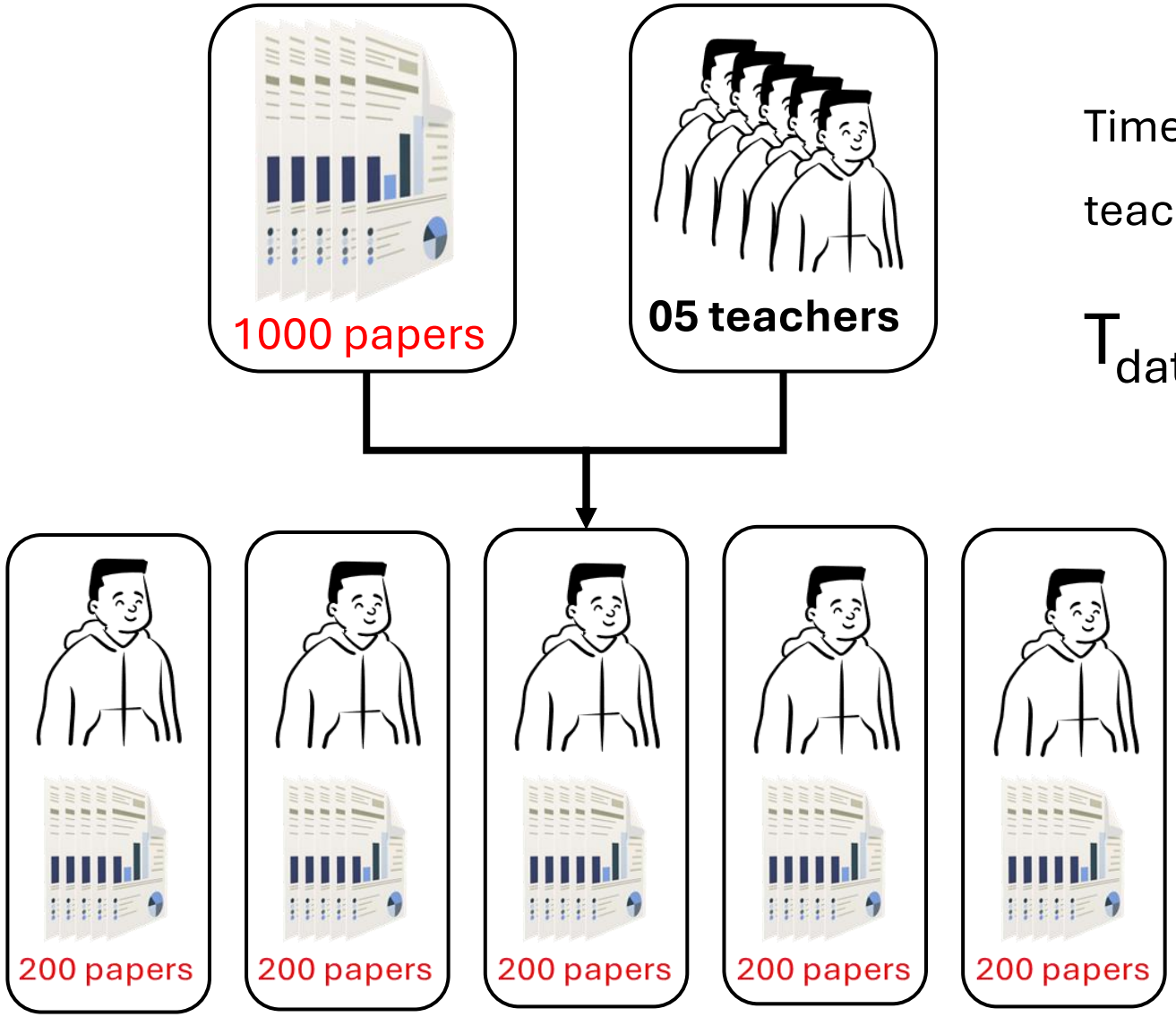
Need 10 minutes to correct a paper for a teacher

Total time to correct whole papers is T_s ,

$$T_s = 1000 \times 10 = 10000 \text{ min}$$

Serial processing

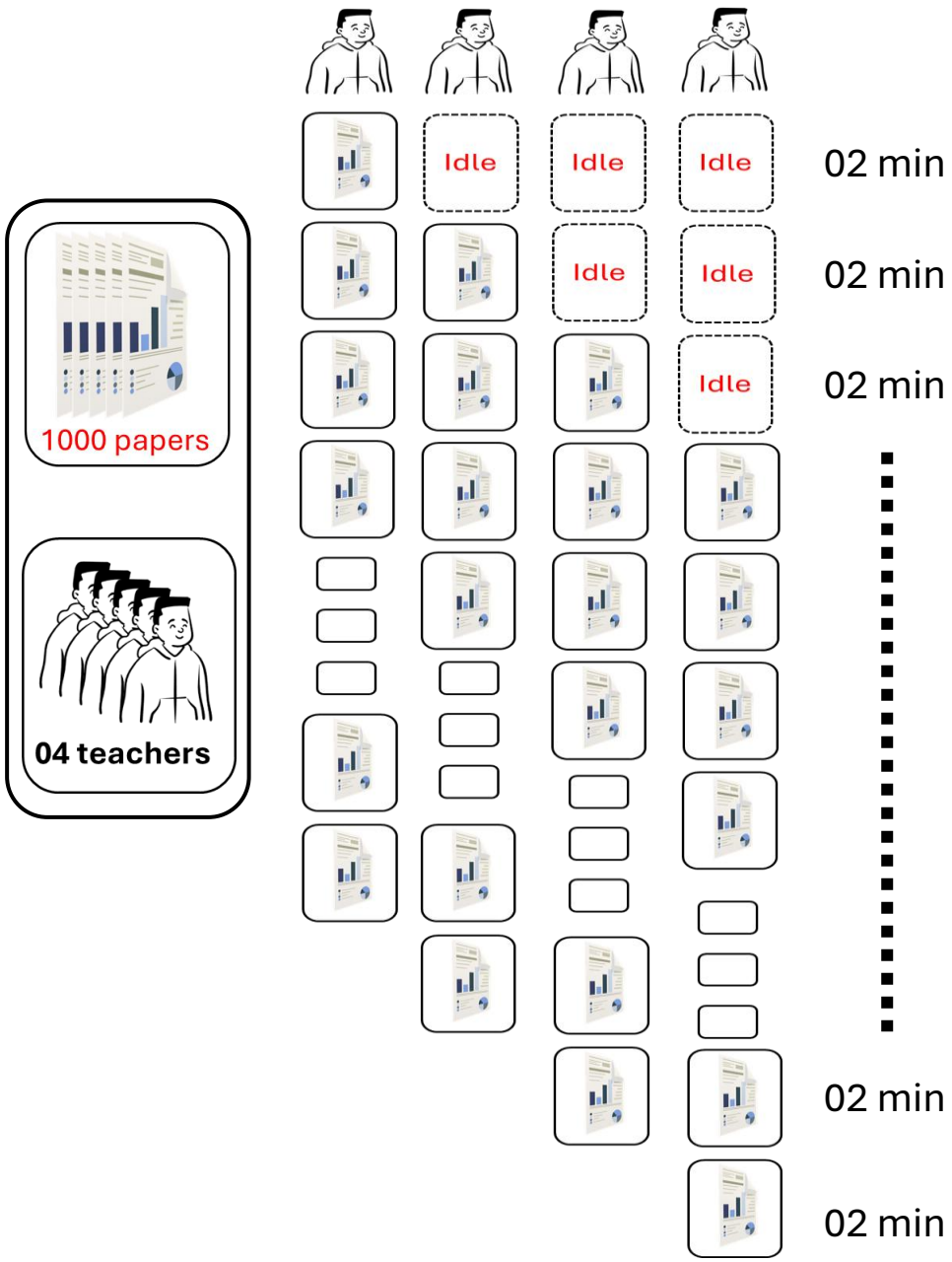
Data-based Parallelism



Time to correct whole papers for all teachers is T_{data_p}

T_{data_p} = 200 x 10 = 2000 min

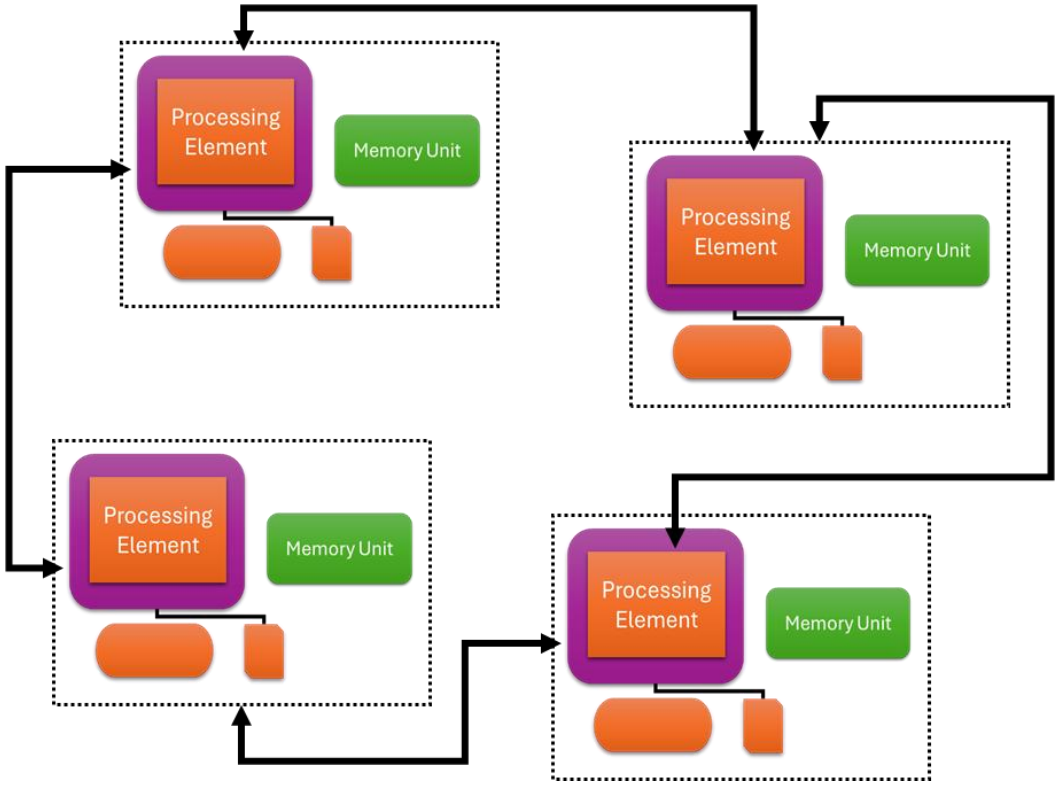
Task-based Parallelism



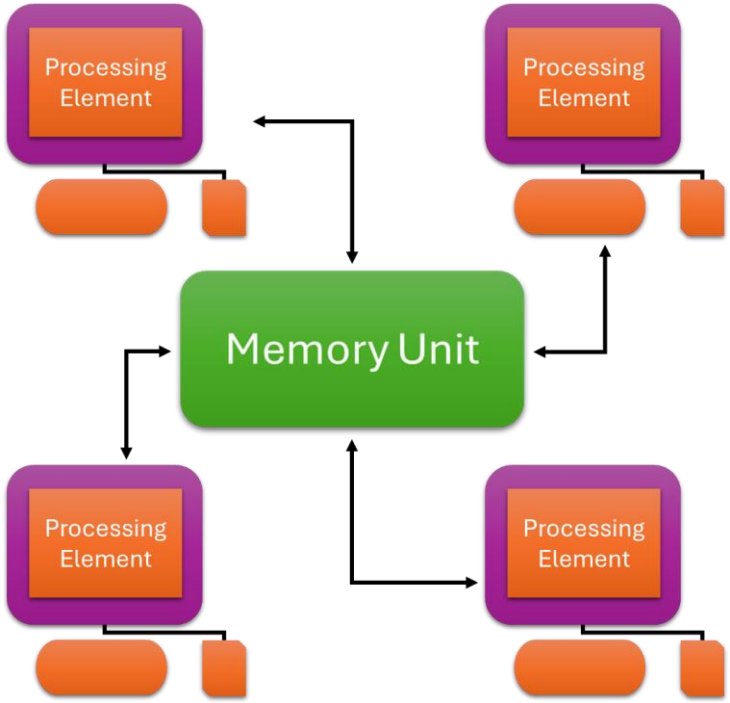
There 4 questions in the paper but the teachers are only experts in each questions and the time to complete those each question is 2 minutes for a teacher.

$T_{task_p} = 1000 \times 2 = 2000$ min

Distributed Memory Systems



Shared Memory Systems



Shared memory → like many people reading and writing on one big whiteboard
Distributed memory → like many people each having their own notebook and passing notes to each other

Distributed Memory Systems

- Each processing elements(PEs) have its own memory for executions.
- Programmer Explicitly writes program for making communication between each PE.
- MPI(Message Passing Interface) is used for writing parallel programming, the core idea of MPI,
  - Every process runs **the same program** (Single Program Multiple Data = SPMD model).
  - Each process gets a unique number called **rank** (0, 1, 2, …, n-1).
  - Processes talk only by sending/receiving messages

Shared Memory Systems

- Each processing elements(PEs) shares a common memory for executions.
- Writing program for shared memory systems is easier than distributed memory systems.
- Communication between each PEs is as much as faster than Distributed memory systems.
- OpenMP and CUDA are mostly used frameworks for writing parallel programming for shared memory systems.